

Validation against held-out adjudicated labels

source: outputs/runs/run-20260922T063622Z/module_b/validation_report.json (research edition)

```
{
  "module_a": {
    "truth_pairs": 95,
    "predicted_pairs": 123,
    "true_positive": 94,
    "false_positive": 29,
    "false_negative": 1,
    "precision": 0.7642,
    "recall": 0.9895,
    "refusal_ratio": 0.0956,
    "baseline_a0_unconstrained": {
      "predicted_pairs": 1164,
      "true_positive": 2,
      "false_positive": 1162,
      "false_negative": 93,
      "precision": 0.0017,
      "recall": 0.0211
    }
  },
  "module_b": {
    "snapshots": 959,
    "test_snapshots": 192,
    "outcomes": {
      "payment": 576,
      "still_open": 375,
      "close_no_payment": 8
    },
    "models": {
      "B0_age_stage_baseline": {
        "status": "FITTED",
        "snap_brier": 0.4943193812002867,
        "snap_log_loss": 0.7148682447611714,
        "n_snapshots": 192,
        "n_abstain": 0
      },
      "B1_discrete_multinomial": {
        "status": "FITTED",
        "snap_brier": 0.4908483777418546,
        "snap_log_loss": 0.7123097346939679,
        "n_snapshots": 192,
        "n_abstain": 0
      },
      "C1_quality_augmented": {
        "status": "FITTED",
        "snap_brier": 0.4914677676105672,
        "snap_log_loss": 0.7124455697623446,
        "n_snapshots": 192,
        "n_abstain": 0
      }
    }
  },
  "ablations": {
```

```
"quality_features": {
  "B1_snap_log_loss": 0.7123097346939679,
  "C1_snap_log_loss": 0.7124455697623446,
  "delta": 0.0001,
  "interpretation": "quality features add no incremental value on the locked test set"
},
"candidate_vs_baseline": {
  "B0_snap_log_loss": 0.7148682447611714,
  "B1_snap_log_loss": 0.7123097346939679,
  "relative_improvement": 0.0036
}
},
"limitations": {
  "study_scope": "All figures come from the study dataset of three participating sites.
No research claim transfers to real operations.",
  "censored_share": 0,
  "close_class_rarity": 8,
  "leakage_audit": "T10/T13/T14/T15 enforced: as-of truncation, forbidden fields,
lineage-grouped splits, train-only preprocessing."
}
}
```